

MATH 3150 — Notes

1 Subsequences

Definition 1.1 (Subsequence). Let $(x_n)_{n=1}^\infty$ be a sequence of real numbers, and let $(n_k)_{k=1}^\infty$ be a sequence of natural numbers that is **strictly increasing**:

$$n_1 < n_2 < n_3 < \cdots.$$

The sequence $(x_{n_k})_{k=1}^\infty = (x_{n_1}, x_{n_2}, x_{n_3}, \dots)$ is called a **subsequence** of (x_n) .

Equivalently, a subsequence is $x \circ \varphi$, where $x : \mathbb{N} \rightarrow \mathbb{R}$ is the sequence and $\varphi : \mathbb{N} \rightarrow \mathbb{N}$, $\varphi(k) = n_k$, is strictly increasing.

Remark 1.2.

- The *index* of a subsequence is k , not n . The k -th term of the subsequence is x_{n_k} .
- A subsequence keeps terms **in their original order** and never repeats an index. You may skip terms, but you may not reorder them or reuse them.
- Every sequence is a subsequence of itself (take $n_k = k$).

Lemma 1.3. If (n_k) is a strictly increasing sequence in $\mathbb{N}$, then $n_k \geq k$ for all $k \in \mathbb{N}$.

Proof. Induction on k . For $k = 1$: $n_1 \in \mathbb{N}$, so $n_1 \geq 1$. If $n_k \geq k$, then $n_{k+1} > n_k \geq k$, and since n_{k+1} is an integer, $n_{k+1} \geq k + 1$. $\square$

Example 1.4. Let $x_n = (-1)^n$. Then

- $n_k = 2k$ gives the subsequence $x_{2k} = 1, 1, 1, \dots$;
- $n_k = 2k - 1$ gives the subsequence $x_{2k-1} = -1, -1, -1, \dots$.

Example 1.5. Let $x_n = \frac{1}{n}$. With $n_k = k^2$, the subsequence is $x_{k^2} = \frac{1}{k^2}$, i.e. $1, \frac{1}{4}, \frac{1}{9}, \dots$.

Example 1.6 (Non-examples). For $x_n = \frac{1}{n}$:

- $(\frac{1}{2}, 1, \frac{1}{4}, \frac{1}{3}, \dots)$ is *not* a subsequence, because the terms are out of order ($n_1 = 2 > n_2 = 1$).
- $(1, 1, \frac{1}{2}, \frac{1}{2}, \dots)$ is *not* a subsequence, because indices repeat ($n_1 = n_2 = 1$).

Theorem 1.7. If $x_n \rightarrow L$, then every subsequence (x_{n_k}) also converges to L .

Proof. Let $\varepsilon > 0$. Since $x_n \rightarrow L$, there is N with $|x_n - L| < \varepsilon$ for all $n \geq N$. If $k \geq N$, then $n_k \geq k \geq N$ by Lemma 1.3, so $|x_{n_k} - L| < \varepsilon$. $\square$

Consequence. If a sequence has two subsequences with different limits, it diverges. For example, $(-1)^n$ diverges because $x_{2k} \rightarrow 1$ and $x_{2k-1} \rightarrow -1$.